Unit 3

Lesson Four

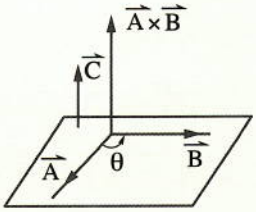
Vector product and scalar triple product

• Vector product of two vectors (Cross product) :

Definition

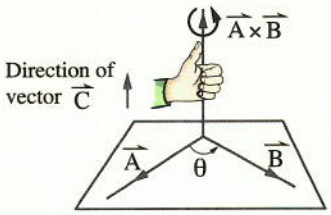
If $\vec{A}$, $\vec{B}$ are two non-zero vectors , θ is the measure of the smaller angle between them when they are drawn from the same starting point , then the vector product of $\vec{A}$ by $\vec{B}$, denoted as $\vec{A} \times \vec{B}$, is defined as $\vec{A} \times \vec{B} = \left(\| \vec{A} \| \| \vec{B} \| \sin \theta \right) \vec{C}$

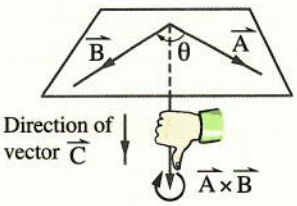
where $\vec{C}$ is a unit vector perpendicular to the plane containing the two vectors $\vec{A}$ and $\vec{B}$ and its direction (up or down) is according to the right hand rule as follows :



The right hand rule :

When the curved fingers of the right hand show the direction of rotation from vector $\vec{A}$ to vector $\vec{B}$ across the smaller angle between them , the thump shows the direction of the vector $\vec{C}$





Remarks

1. If $\vec{A}$, $\vec{B}$ are two non-zero vectors , θ is the measure of the smaller angle between them , then :

   • If $\vec{A} \times \vec{B}$ is in direction of unit vector $\vec{C}$, then $\vec{B} \times \vec{A}$ is in direction of unit vector $-\vec{C}$

   i.e. $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$

   • The vector product $\vec{A} \times \vec{B}$ is a vector , not scalar as in the scalar product.

- $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$ but $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$
- $\| \vec{A} \times \vec{B} \| = \| \vec{A} \| \| \vec{B} \| \sin\theta$, $\sin\theta = \dfrac{\| \vec{A} \times \vec{B} \|}{\| \vec{A} \| \| \vec{B} \|}$
- The unit vector in direction of $\vec{A} \times \vec{B}$ is $\vec{C} = \dfrac{\vec{A} \times \vec{B}}{\| \vec{A} \times \vec{B} \|} = \dfrac{\vec{A} \times \vec{B}}{\| \vec{A} \| \| \vec{B} \| \sin\theta}$

2. If $\vec{A}$, $\vec{B}$, $\vec{C}$ are three non-zero vectors and $\vec{A} \times \vec{B} = \vec{A} \times \vec{C}$, then it is not necessary that $\vec{B} = \vec{C}$

• Properties of the vector product of two vectors :

If $\vec{A}$ and $\vec{B}$ are two vectors, θ is the measure of the angle between them , then :

1. $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$ (the cross product is not commutative)

2. $\vec{A} \times (\vec{B} + \vec{C}) = (\vec{A} \times \vec{B}) + (\vec{A} \times \vec{C})$

, $(\vec{B} + \vec{C}) \times \vec{A} = \vec{B} \times \vec{A} + \vec{C} \times \vec{A}$ (distribution property)

3. $(k\,\vec{A}) \times \vec{B} = \vec{A} \times (k\,\vec{B}) = k\,(\vec{A} \times \vec{B})$ where k is a real number.

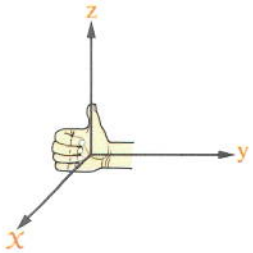
4. $\vec{A} \times \vec{A} = \vec{B} \times \vec{B} = \vec{O}$

i.e. (The result of vector product of any vector by itself $= \vec{O}$, where $\theta = 0$)

5. If $\vec{A} \times \vec{B} = \vec{O}$, then either $\vec{A} \,//\, \vec{B}$ or one of them or both is equal to $\vec{O}$

• The vector product of the fundamental unit vectors of a right hand system :

If $\{\hat{i}$, $\hat{j}$, $\hat{k}\}$ is a set of right hand system of unit vectors , then :



- $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{O}$
- $\hat{i} \times \hat{j} = \hat{k}$

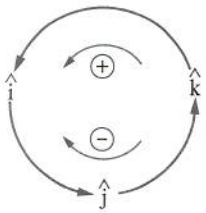
, $\hat{j} \times \hat{k} = \hat{i}$

, $\hat{k} \times \hat{i} = \hat{j}$

- $\hat{j} \times \hat{i} = -\hat{k}$

, $\hat{k} \times \hat{j} = -\hat{i}$

, $\hat{i} \times \hat{k} = -\hat{j}$



• The vector product of two vectors in the cartesian form :

If $\vec{A} = (A_x , A_y , A_z)$, $\vec{B} = (B_x , B_y , B_z)$ are two non-zero vectors , then :

$$\vec{A} \times \vec{B} = (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$$

$$\therefore \vec{A} \times \vec{B} = A_x \hat{i} \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) + A_y \hat{j} \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$$
$$+ A_z \hat{k} \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$$

$$\therefore \vec{A} \times \vec{B} = A_x B_x (\hat{i} \times \hat{i}) + A_x B_y (\hat{i} \times \hat{j}) + A_x B_z (\hat{i} \times \hat{k}) + A_y B_x (\hat{j} \times \hat{i}) + A_y B_y (\hat{j} \times \hat{j})$$
$$+ A_y B_z (\hat{j} \times \hat{k}) + A_z B_x (\hat{k} \times \hat{i}) + A_z B_y (\hat{k} \times \hat{j}) + A_z B_z (\hat{k} \times \hat{k})$$

$$\therefore \vec{A} \times \vec{B} = \vec{O} + A_x B_y (\hat{k}) + A_x B_z (-\hat{j}) + A_y B_x (-\hat{k}) + \vec{O} + A_y B_z (\hat{i}) + A_z B_x \hat{j}$$
$$+ A_z B_y (-\hat{i}) + \vec{O}$$

$$\therefore \vec{A} \times \vec{B} = (A_y B_z - A_z B_y) \hat{i} + (A_z B_x - A_x B_z) \hat{j} + (A_x B_y - A_y B_x) \hat{k}$$

and the last form can be written in 3×3 determinant form as follows $\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$

• Special case :

If $\vec{A} = (A_x , A_y)$, $\vec{B} = (B_x , B_y)$ in the x y-plane , then :

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & 0 \\ B_x & B_y & 0 \end{vmatrix} = (A_x B_y - A_y B_x) \hat{k}$$

and in an another form $\vec{A} \times \vec{B} = \begin{vmatrix} A_x & A_y \\ B_x & B_y \end{vmatrix} \hat{k} = (A_x B_y - A_y B_x) \hat{k}$

and also $\vec{A} \times \vec{B} = (A_x , A_y) \times (B_x , B_y) = (A_x B_y - A_y B_x) \hat{k}$

Example 1

If $\|\vec{A}\| = 5$, $\|\vec{B}\| = 8$, $\theta = 30°$, find : $\|\vec{A} \times \vec{B}\|$

Solution

$$\|\vec{A} \times \vec{B}\| = \|\vec{A}\| \, \|\vec{B}\| \sin\theta = 5 \times 8 \times \sin 30° = 20$$

Example 2

If $\vec{A} \times \vec{B} = -35\sqrt{3}\,\vec{C}$, $\|\vec{A}\| = 14$, $\|\vec{B}\| = 5$, find the measure of the angle between the two vectors $\vec{A}$, $\vec{B}$

Solution

$$\because \sin\theta = \frac{\|\vec{A} \times \vec{B}\|}{\|\vec{A}\|\,\|\vec{B}\|} = \frac{35\sqrt{3}}{14 \times 5} = \frac{\sqrt{3}}{2}$$

$$\therefore \theta = 60° \text{ or } 120°$$

Example 3

If $\vec{A} = (1 , 2 , 1)$, $\vec{B} = 3\hat{i} + \hat{j} + 2\hat{k}$, find :

1. $\vec{A} \times \vec{B}$ **2.** $\vec{B} \times \vec{A}$

3. Unit vector perpendicular to each of the two vectors $\vec{A}$, $\vec{B}$

Solution

1. $$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 3 & 1 & 2 \end{vmatrix}$$

$$= (2 \times 2 - 1 \times 1)\hat{i} - (1 \times 2 - 3 \times 1)\hat{j} + (1 \times 1 - 3 \times 2)\hat{k} = 3\hat{i} + \hat{j} - 5\hat{k}$$

2. $$\vec{B} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 2 \\ 1 & 2 & 1 \end{vmatrix}$$

Notice that
$\vec{B} \times \vec{A} = -\vec{A} \times \vec{B}$

$$= (1 \times 1 - 2 \times 2)\hat{i} - (3 \times 1 - 2 \times 1)\hat{j} + (3 \times 2 - 1 \times 1)\hat{k} = -3\hat{i} - \hat{j} + 5\hat{k}$$

3. There are two unit vectors perpendicular to the two vectors $\vec{A}$, $\vec{B}$

$$\frac{\vec{A} \times \vec{B}}{\|\vec{A} \times \vec{B}\|} = \frac{3\hat{i} + \hat{j} - 5\hat{k}}{\sqrt{9 + 1 + 25}} = \frac{3}{\sqrt{35}}\hat{i} + \frac{1}{\sqrt{35}}\hat{j} - \frac{5}{\sqrt{35}}\hat{k}$$

$$\text{or } \frac{\vec{B} \times \vec{A}}{\|\vec{B} \times \vec{A}\|} = \frac{-3\hat{i} - \hat{j} + 5\hat{k}}{\sqrt{35}} = \frac{-3}{\sqrt{35}}\hat{i} - \frac{1}{\sqrt{35}}\hat{j} + \frac{5}{\sqrt{35}}\hat{k}$$

i.e. The two vectors are $\pm\left(\frac{3}{\sqrt{35}}\hat{i} + \frac{1}{\sqrt{35}}\hat{j} - \frac{5}{\sqrt{35}}\hat{k}\right)$

Parallelism of two vectors

When two vectors are parallel , then the measure of the angle between them is either :

zero if they are in the same direction or 180° if they are in opposite directions

, $\because \vec{A} \times \vec{B} = \left(\| \vec{A} \| \| \vec{B} \| \sin\theta\right) \vec{C}$, and since $\sin 0° = \sin 180° = 0$

$\therefore$ In case of parallelism $\vec{A} \times \vec{B} = \vec{O}$

, from the vector product definition in cartesian coordinate

$\therefore (A_y B_z - A_z B_y)\hat{i} + (A_z B_x - A_x B_z)\hat{j} + (A_x B_y - A_y B_x)\hat{k} = \vec{O}$

$\therefore A_y B_z = A_z B_y$, and so $\frac{A_y}{B_y} = \frac{A_z}{B_z}$ (1)

, $A_z B_x = A_x B_z$, and so $\frac{A_z}{B_z} = \frac{A_x}{B_x}$ (2)

, $A_x B_y = A_y B_x$, and so $\frac{A_x}{B_x} = \frac{A_y}{B_y}$ (3)

From (1) , (2) , (3) :

$\therefore \frac{A_x}{B_x} = \frac{A_y}{B_y} = \frac{A_z}{B_z}$ and let each of the ratios = k

$\therefore A_x = k B_x$, $A_y = k B_y$, $A_z = k B_z$

, $\because \vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$

$\therefore \vec{A} = k B_x \hat{i} + k B_y \hat{j} + k B_z \hat{k} = k (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$

$\therefore \vec{A} = k \vec{B}$ $\therefore \vec{A} // \vec{B}$

, in case $k > 0$, then $\vec{A} // \vec{B}$ and in the same direction.

, in case $k < 0$, then $\vec{A} // \vec{B}$ and in the opposite direction of each other.

So

If $\vec{A}$, $\vec{B}$ are two non-zero vectors and

- $\vec{A} \times \vec{B} = \vec{O}$, then $\vec{A} // \vec{B}$ and vice versa.
- $\frac{A_x}{B_x} = \frac{A_y}{B_y} = \frac{A_z}{B_z}$, then $\vec{A} // \vec{B}$ and vice versa.

Example 4

Let the vector $\vec{A} = 5\hat{i} - 4\hat{j} + m\hat{k}$, $\vec{B} = 6\hat{i} + \ell\hat{j} - 12\hat{k}$ and $\vec{A} \,//\, \vec{B}$ Find each of ℓ and m

Solution

$\because \vec{A} \,//\, \vec{B}$ $\quad\quad$ $\therefore \frac{A_x}{B_x} = \frac{A_y}{B_y} = \frac{A_z}{B_z}$

$\therefore \frac{5}{6} = \frac{-4}{\ell} = \frac{m}{-12}$ $\quad\quad$ $\therefore \ell = \frac{-4 \times 6}{5} = -4.8$, $m = \frac{5 \times -12}{6} = -10$

Example 5

If $\vec{A} = (-1, 2, 2)$, $\|\vec{B}\| = 6$, $\vec{A} \,//\, \vec{B}$, find : $\vec{B}$

Solution

$\because \vec{A} \,//\, \vec{B}$ $\quad\quad$ $\therefore \vec{B} = k\vec{A}$

, $\because \|\vec{B}\| = |k| \|\vec{A}\|$ $\quad\quad$ $\therefore 6 = |k| \sqrt{(-1)^2 + (2)^2 + (2)^2}$

$\therefore |k| = 2$ $\quad\quad$ $\therefore k = \pm 2$

$\therefore \vec{B} = 2(-1, 2, 2) = (-2, 4, 4)$ $\quad\quad$ Or $\vec{B} = -2(-1, 2, 2) = (2, -4, -4)$

Remark

If A , B , C are three points in 3D-space and

1. $\overrightarrow{AB} \times \overrightarrow{BC} = \vec{O}$, then A , B , C are collinear.
2. $\overrightarrow{AB} \,//\, \overrightarrow{BC}$, then A , B , C are collinear.

Example 6

Prove that : A , B , C are collinear where A (1 , 2 , 3) , B (4 , 0 , 4) , C (– 2 , 4 , 2)

Solution

$\overrightarrow{AB} = \vec{B} - \vec{A} = (4, 0, 4) - (1, 2, 3) = (3, -2, 1)$

, $\overrightarrow{BC} = \vec{C} - \vec{B} = (-2, 4, 2) - (4, 0, 4) = (-6, 4, -2)$

$$\therefore \overrightarrow{AB} \times \overrightarrow{BC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & 1 \\ -6 & 4 & -2 \end{vmatrix}$$

$$= (-2 \times -2 - 4 \times 1)\hat{i} - (3 \times -2 - 1 \times -6)\hat{j} + \left(3 \times 4 - (-2) \times (-6)\right)\hat{k}$$

$$= (4 - 4)\hat{i} - (-6 + 6)\hat{j} + (12 - 12)\hat{k} = \vec{O}$$

$\therefore$ A , B , C are collinear.

The geometrical using of the vector product of two vectors

If $\vec{A}$ and $\vec{B}$ are two non parallel vectors on the same plane , then :

$$\| \vec{A} \times \vec{B} \| = \| \vec{A} \| \| \vec{B} \| \sin\theta$$

«where θ is the measure of the included angle between $\vec{A}$ and $\vec{B}$ »

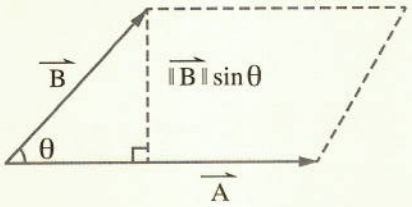
, from the previous we deduce the following :

1. Area of parallelogram in which $\vec{A}$ and $\vec{B}$ are to adjacent sides

$= \text{base length} \times \text{corresponding height}$

$= \| \vec{A} \| \times \| \vec{B} \| \sin\theta$

$= \| \vec{A} \times \vec{B} \| =$ Norm of the vector product of the two vectors



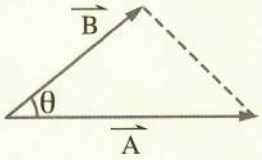
2. Area of triangle in which $\vec{A}$ and $\vec{B}$ are two adjacent sides

$= \frac{1}{2}$ product of lengths of two adjacent sides × sine of included angle between them

$= \frac{1}{2} \| \vec{A} \| \times \| \vec{B} \| \sin\theta$

$= \frac{1}{2} \| \vec{A} \times \vec{B} \|$

$= \frac{1}{2}$ Norm of the vector product of the two vectors



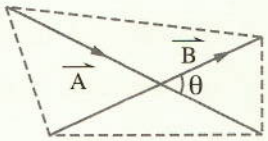
3. Area of quadrilateral in which $\vec{A}$ and $\vec{B}$ are two diagonals

$= \frac{1}{2}$ product of the lengths of the two diagonals × sine the included angle between them

$= \frac{1}{2} \| \vec{A} \| \times \| \vec{B} \| \sin\theta$

$= \frac{1}{2} \| \vec{A} \times \vec{B} \|$

$= \frac{1}{2}$ Norm of the vector product of the two vectors



Example 7

Find the area of the parallelogram in which $\vec{A}$, $\vec{B}$ are two adjacent sides where

$\vec{A} = (3 , 6 , 3)$, $\vec{B} = (-6 , -2 , -4)$

Solution

$$\because \vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 6 & 3 \\ -6 & -2 & -4 \end{vmatrix} = -18\hat{i} - 6\hat{j} + 30\hat{k}$$

$\therefore$ The area of the parallelogram $= \| \vec{A} \times \vec{B} \| = \sqrt{(-18)^2 + (-6)^2 + (30)^2} = 6\sqrt{35}$ square units.

The scalar triple product

If $\vec{A}$, $\vec{B}$, $\vec{C}$ are vectors in 3D-space , then the scalar product of the two vectors $\vec{A}$, $(\vec{B} \times \vec{C})$ is known as the scalar triple product of the vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ and written as $\vec{A} \cdot (\vec{B} \times \vec{C})$ and since doing the scalar product first is meaningless , the brackets can be neglected and the scalar triple product could be written as $\vec{A} \cdot \vec{B} \times \vec{C}$

Let $\vec{A} = (A_x , A_y , A_z)$, $\vec{B} = (B_x , B_y , B_z)$, $\vec{C} = (C_x , C_y , C_z)$

$$\therefore \vec{A} \cdot \vec{B} \times \vec{C} = (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \cdot \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

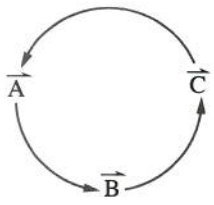
$$= (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \cdot \left[\begin{vmatrix} B_y & B_z \\ C_y & C_z \end{vmatrix} \hat{i} - \begin{vmatrix} B_x & B_z \\ C_x & C_z \end{vmatrix} \hat{j} + \begin{vmatrix} B_x & B_y \\ C_x & C_y \end{vmatrix} \hat{k} \right]$$

$$= A_x (B_y C_z - B_z C_y) - A_y (B_x C_z - B_z C_x) + A_z (B_x C_y - B_y C_x)$$

$$\therefore \vec{A} \cdot \vec{B} \times \vec{C} = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

• The properties of the scalar triple product :

1. The value of the scalar triple product does not change if the vectors are permuted in such a way that they are still read in the same cyclic order.



$\vec{A} \cdot \vec{B} \times \vec{C} = \vec{B} \cdot \vec{C} \times \vec{A} = \vec{C} \cdot \vec{A} \times \vec{B}$

2. The symbols of the scalar product and the vector product can be switched preserving the cyclic order of the vectors and the result of the scalar triple product will not change.

i.e. $\vec{A} \cdot \vec{B} \times \vec{C} = \vec{A} \times \vec{B} \cdot \vec{C}$

3. If the vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ are in the same plane , then the scalar triple product vanishes.

i.e. $\vec{A} \cdot \vec{B} \times \vec{C} = 0$

Notice that

If $\vec{A} \cdot \vec{B} \times \vec{C}$ = zero , then either $\vec{A}$, $\vec{B}$, $\vec{C}$ are in the same plane

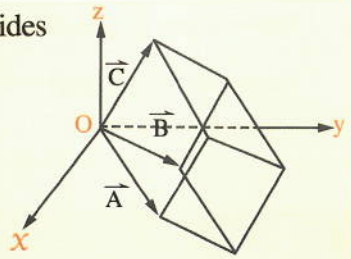
Or $\vec{A}$, $\vec{B}$, $\vec{C}$ are parallel.

4. If $\overrightarrow{AB} \cdot \overrightarrow{AC} \times \overrightarrow{AD} = 0$, then the points A , B , C , D are in the same plane.

The geometrical using of the scalar triple product

If $\vec{A}$, $\vec{B}$, $\vec{C}$ are three vectors forming 3-non parallel connected sides of a parallelepiped , then the volume of the parallelpiped = the absolute value of the scalar triple product of these vectors.

Thus the volume of the parallelepiped = $|\vec{A} \cdot \vec{B} \times \vec{C}|$



Notice that

The parallelepiped is a solid generated by translating the surface of a parallelogram parallel to itself in a fixed direction.

So there are six faces in the parallelepiped each of them is a parallelogram and each two opposite faces are parallel and congruent and in case :

1. The parallel bases are parallelograms and the lateral faces are rectangles , then it is called a right parallelepiped.
2. The six faces are rectangles , it is called a cuboid.
3. The six faces are squares , it is called a cube.

Example 8

Find the volume of the parallelepiped in which three adjacent sides are represented by the vectors $\vec{A} = (1 , 2 , -1)$, $\vec{B} = (-1 , 2 , 2)$, $\vec{C} = (2 , 0 , 1)$

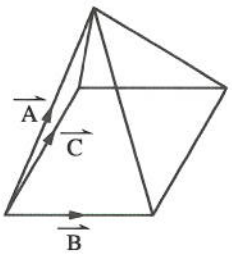
Solution

$$\because \vec{A} \cdot \vec{B} \times \vec{C} = \begin{vmatrix} 1 & 2 & -1 \\ -1 & 2 & 2 \\ 2 & 0 & 1 \end{vmatrix} = 1\,(2-0) - 2\,(-1-4) - 1\,(0-4) = 16$$

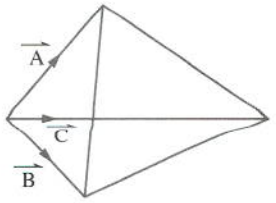
$\therefore$ The volume of the parallelepiped $= |\vec{A} \cdot \vec{B} \times \vec{C}| = |16| = 16$ volume unit.

Remarks

- The height of the parallelepiped corresponding to the base formed from the two vectors $\vec{B}$, $\vec{C}$ equals $\dfrac{|\vec{A} \cdot \vec{B} \times \vec{C}|}{\|\vec{B} \times \vec{C}\|}$
- Volume of the quadrilateral pyramid whose base is a parallelogram in which the two vectors $\vec{B}$, $\vec{C}$ are two adjacent sides and the vector $\vec{A}$ is one of the lateral edges equals $\frac{1}{3} |\vec{A} \cdot \vec{B} \times \vec{C}|$



- Volume of the triangular pyramid in which the vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ represents three non coplanar edges of the pyramid equals $\frac{1}{6} | \vec{A} . \vec{B} \times \vec{C} |$



Comparison between the scalar product and the vector product

The scalar product	The vector product
$\vec{A} . \vec{B} = \|\vec{A}\| \|\vec{B}\| \cos\theta$ (scalar quantity)	$\vec{A} \times \vec{B} = \left(\|\vec{A}\| \|\vec{B}\| \sin\theta \right) \vec{C}$ (vector quantity) where $\vec{C}$ is a perpendicular unit vector to the plane containing $\vec{A}$ and $\vec{B}$ and in direction shown by the right hand rule.
$\vec{A} . \vec{B} = A_x B_x + A_y B_y + A_z B_z$	$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$
$\vec{A} . \vec{B} = \vec{B} . \vec{A}$	$\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$
$\vec{A} . \vec{B} =$ zero , $\vec{A}$ and $\vec{B}$ are non-zero vectors , then $\vec{A}$ and $\vec{B}$ are perpendicular.	$\vec{A} \times \vec{B} = \vec{O}$, $\vec{A}$ and $\vec{B}$ are non-zero vectors , then $\vec{A}$ and $\vec{B}$ are parallel.
$\vec{A} . \vec{A} = \|\vec{A}\|^2$	$\vec{A} \times \vec{A} = \vec{O}$
• $\hat{i} . \hat{i} = \hat{j} . \hat{j} = \hat{k} . \hat{k} = 1$ • $\hat{i} . \hat{j} = \hat{j} . \hat{k} = \hat{k} . \hat{i} = \hat{i} . \hat{k} = \hat{j} . \hat{i} = \hat{k} . \hat{j} = 0$	• $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{O}$ • $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$
$\vec{A} . \vec{O} = \vec{O} . \vec{A} = \vec{O} . \vec{O} = 0$	$\vec{A} \times \vec{O} = \vec{O} \times \vec{A} = \vec{O} \times \vec{O} = \vec{O}$
$(m \vec{A}) . \vec{B} = \vec{A} . (m \vec{B}) = m (\vec{A} . \vec{B})$	$(m \vec{A}) \times \vec{B} = \vec{A} \times (m \vec{B}) = m (\vec{A} \times \vec{B})$
$(\vec{A} + \vec{B}) . \vec{C} = \vec{A} . \vec{C} + \vec{B} . \vec{C}$	$(\vec{A} + \vec{B}) \times \vec{C} = \vec{A} \times \vec{C} + \vec{B} \times \vec{C}$